



CSE 4205

Digital Logic Design

Sequential Logic

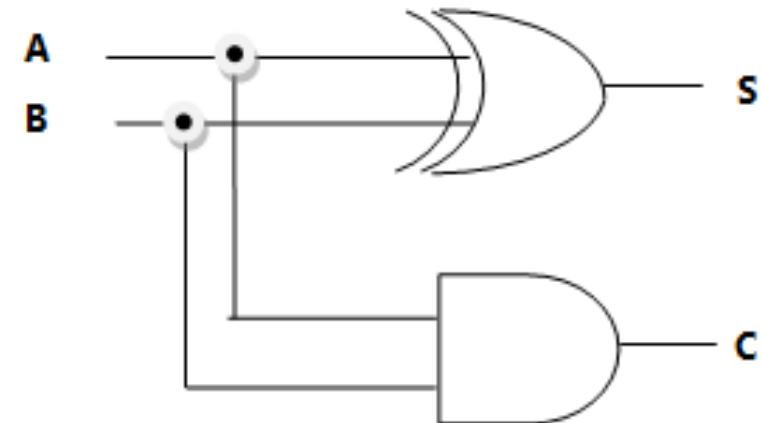
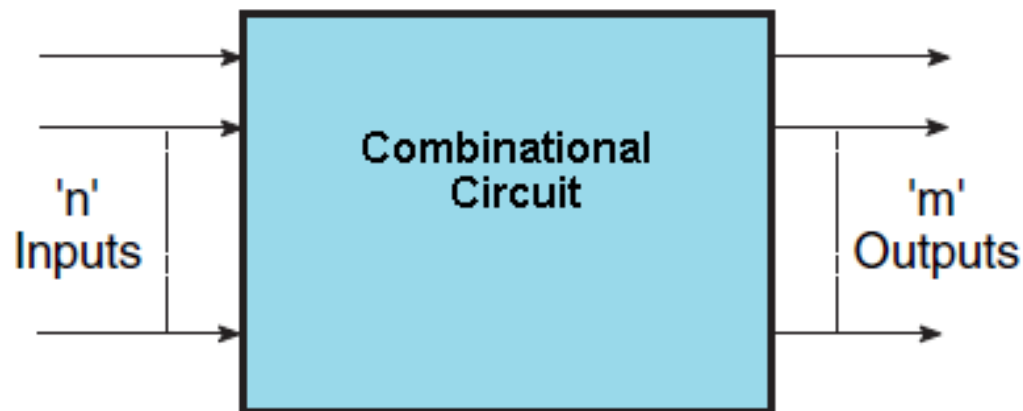
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Introduction – Combinational Circuit

- So far digital circuits considered earlier – **combinational circuit** – **output** at any instant entirely dependent on the **inputs** presents at that time.
- ***Present Output = $f(\text{present inputs})$***
- Example: Adder



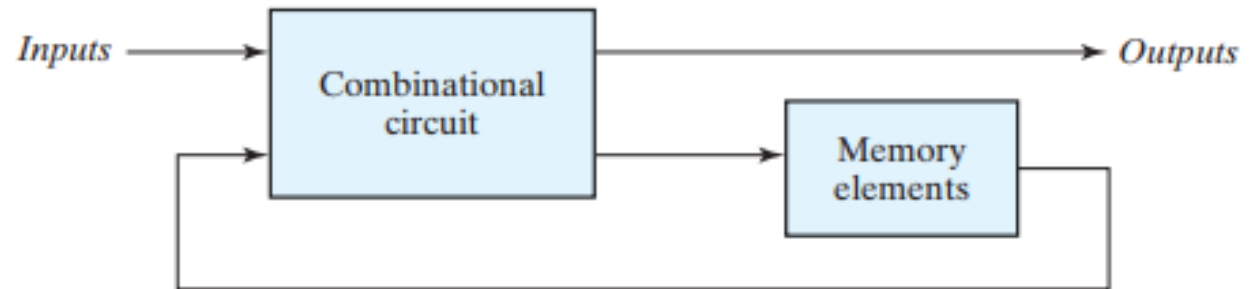
Introduction – Memory

- **Electronic devices** nowadays have the ability to send, receive, store, retrieve and process information in binary format – only possible because of ability to store information (**memory**)
- The circuits that include memory element– **sequential circuit/ logic**
- **Memory elements** are connected to form a feedback path



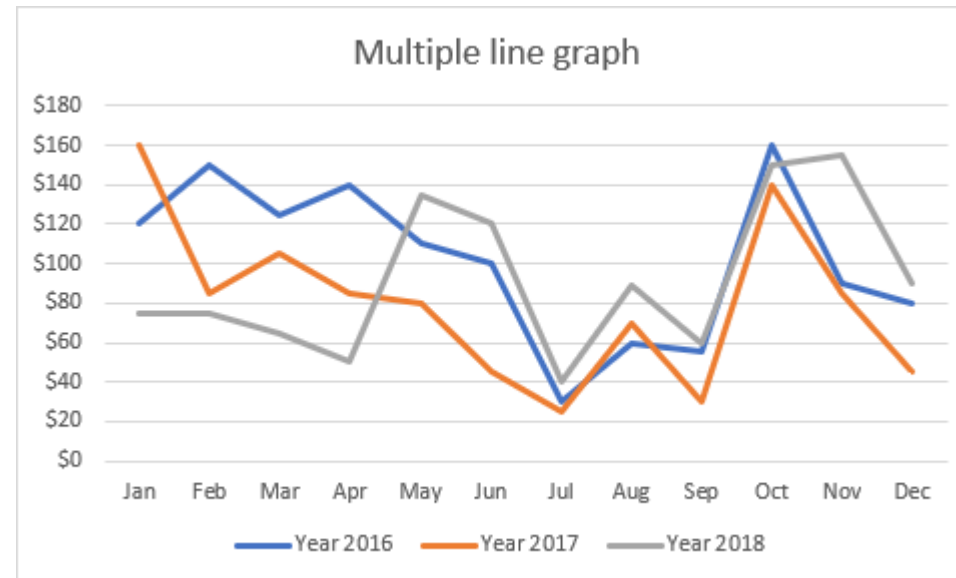
Sequential Circuit

- Combinational circuit + storage elements = Sequential circuit
- Storage elements **insert input** to the combinational circuit through **feedback path**
- **Storage elements** stores binary information – **state (time variant)**
- ***Present output*** = F_1 (*present input, present state*)
- ***Next state of memory element*** = F_2 (*present input, present state*)
- Example : Counter, Register



Time Sequence

- Sequential circuit – **time sequence** of inputs, outputs and internal states

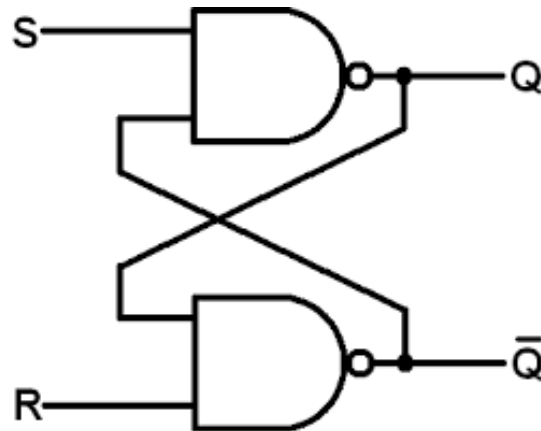


Classification

- *Two types sequential circuit* – based on **timing** of their signal

□ Asynchronous sequential circuit

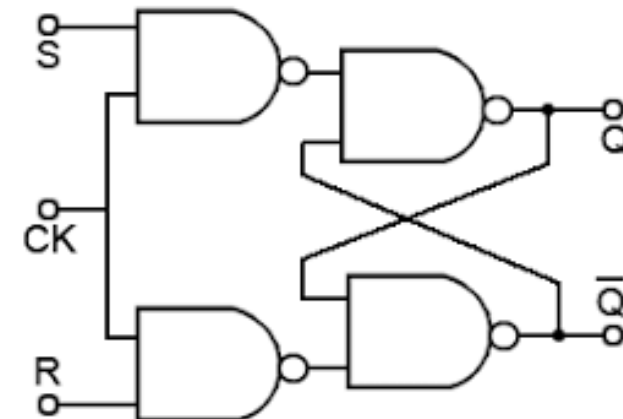
- Behavior defined based on the **order** of input signals at **any instant** of time
- Example: Latch
- **Time delayed device** – propagation delay is higher to provide storage attributes (intentionally with delay units)
- **Drawback:** unstable states at times due to some undesirable feedbacks



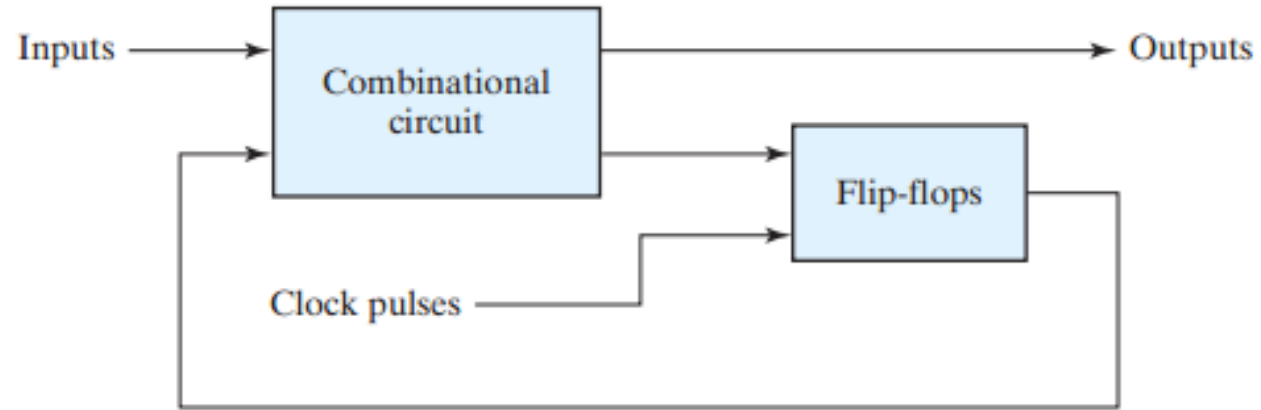
Classification...

□ Synchronous sequential circuit

- Behavior defined based on the signal at **discrete** instants of time
- Example: Flip flop
- Synchronization achieved through timing device – clock generator
- Clock pulse driven storage – only affected with arrival of each pulse
- Clock pulse (clk) – when changes will occur
- External input – what changes will occur
- Benefit – conquer instability problem
- Common practice



Clocked Sequential Circuit

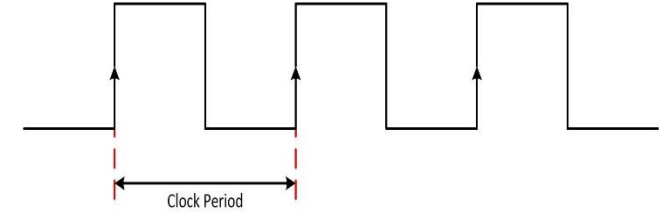


(a) Block diagram

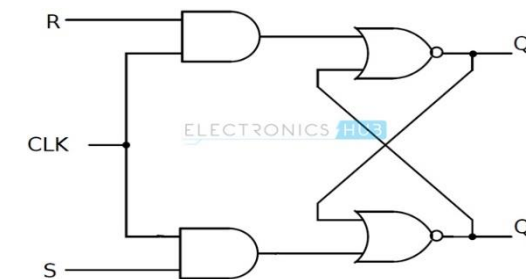


(b) Timing diagram of clock pulses

Clocked Sequential Circuit...



- Synchronous sequential circuit – employs signals **at discrete instants of time** – use pulses of limited duration (pulse amplitude – 0 and 1)
- Inputs from different sources – **have separate independent sources of pulse** – **arrival time may be delayed** (unpredictable) – separate the pulses slightly (not synchronized) – **unreliable operation**
- Binary signal – **voltage amplitude**
- Synchronization achieved through a timing device – followed by all components (synchronized) – **master clock generator** (train of clock pulses) – clock pulse ANDed with specific signal
- **Clocked Sequential circuit**
 - Most frequently used
 - No instability problem





Latch

- Latches are basic storage elements that operate with signal levels (rather than signal transitions).

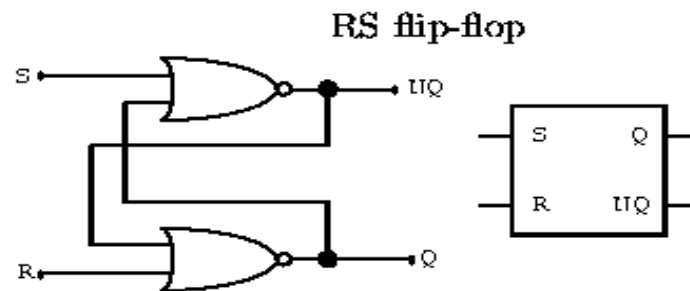


Flip flops

- **Memory elements** in clocked sequential circuit – **flip-flop** – binary cell that stores one bit of information
- Flip-flop – **two outputs** – normal and complement value of the bit stored
- Flip flop – **maintains binary state** indefinitely (based on power supply)
- Differences in flip-flops – based on number of inputs and the manner in which the inputs affect the binary state

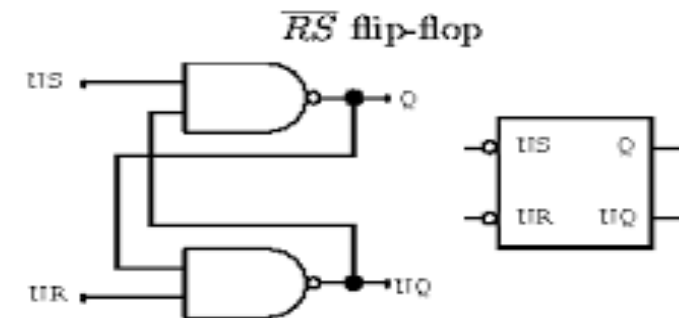
Basic Flip-flop Circuit

- Constructed from NAND and NOR gates which form basic flip-flops
- Two inputs – Reset and Set; Two outputs – Q and Q'
- Q=1, Q'=0 : set (1) state and Q=0, Q'=1 : reset (0) state
- Both inputs should remain in “no change” state unless the state has to be changed and “undefined” state should be avoided



S	R	Q	Q'
0	0	no change	
0	1	0	1
1	0	1	0
1	1	undefined	

Controlled by 1,
if any input= 1,
output =0:
NOR TT



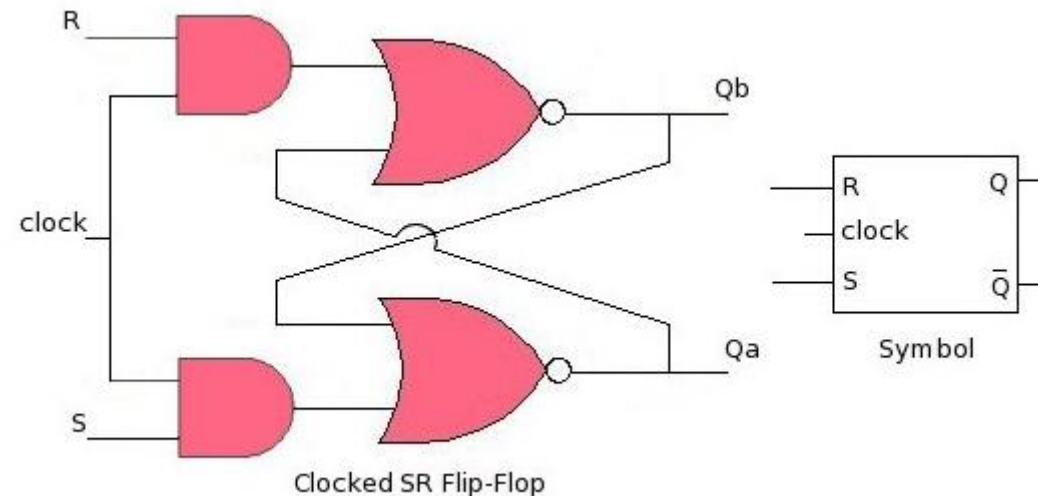
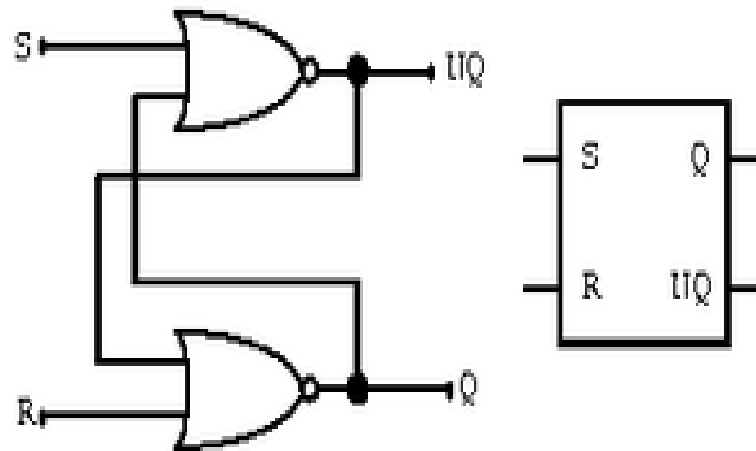
S	R	Q	Q'
0	0	undefined	
0	1	1	0
1	0	0	1
1	1	no change	

Controlled by 0,
if any input= 0,
output =1:
NAND TT

Flip flop and Latches

A **latch** is asynchronous, and the outputs can change as soon as the inputs.

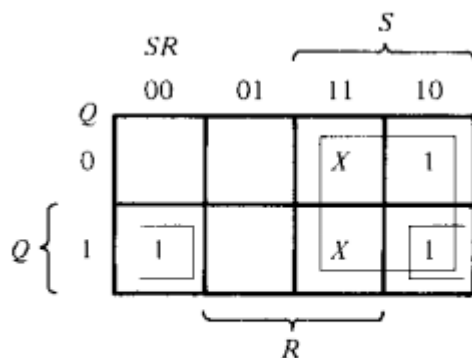
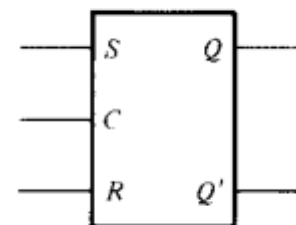
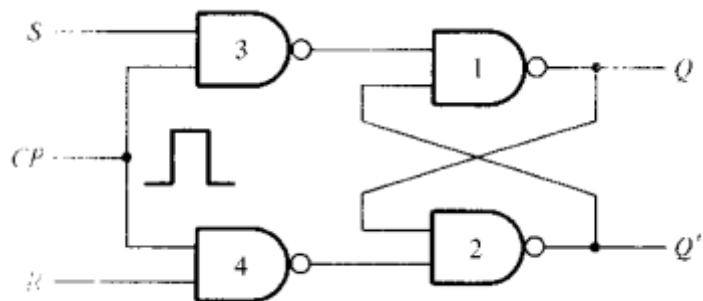
A **flip-flop**, on the other hand, is edge-triggered and only changes state when a control signal goes from high to low or low to high.



RS flip-flop

- **Clocked circuit** – flip-flop will response when it is aligned with clock pulse
- **Clock pulse** – remarked as a triangle in the graphic symbol – Clocks are dynamic indicators – inputs will be responsive at clock's transitions
- **Q** = present state and **Q (t+1)** = next state
- **Characteristic table** summarize the operation of the flip-flop. (R, S, Q – i/p)
- **Graphic symbol** has three inputs (R,S,CP) and two outputs (Q and Q')
- **Characteristic equation** specifies the value of the next state as function of present state and inputs derived from the characteristic table. (R, S, Q – i/p)
- **An extra expression** is also needed to be constrained for indeterminate state (don't care in k-map)

RS flip-flop...



$$Q(t+1) = S + R'Q$$

$$SR = 0$$

En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

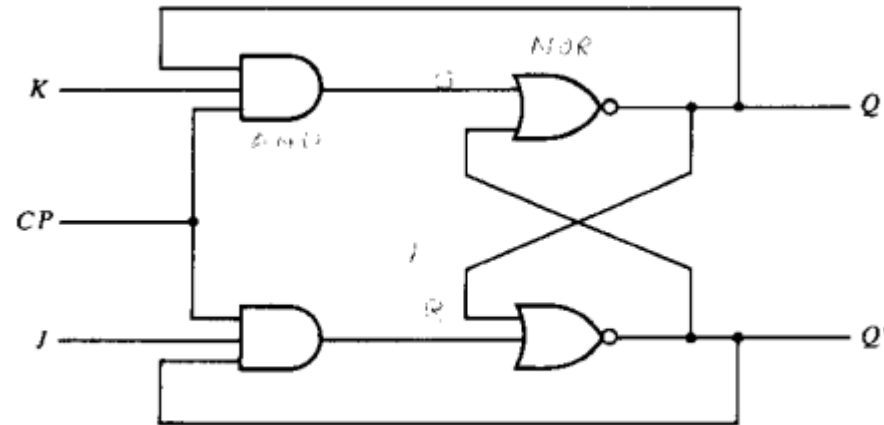
Q	S	R	Q (t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Indeterminate



JK flip-flop

- A **refinement of the RS** flip-flop to avoid indeterminate state
- **J behaves like set (S)** and **K behaves like reset/clear (R)**
- Instead of being indeterminate state it work as **toggle/complement** when both J and K are given input (1) – Q to Q'.
- **Q is ANDed with K and CP** and **Q' is ANDed with J and CP**, it ensures that it will be **cleared if previously set** and **set if previously clear**.
- JK flip-flop behaves like a RS flip-flop *except both of them are 1*.
- *When both of them are 1*, it will allow **one AND gate** to transmit which is connected to the “**current memory state=1**”
- **When CP=J=K=1**, causes repeated and continuous transitions of the output (Race around condition)

JK flip-flop...



(a) Logic diagram

Q	J	K	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Flip-Flop Characteristic Tables

JK Flip-Flop			
J	K	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

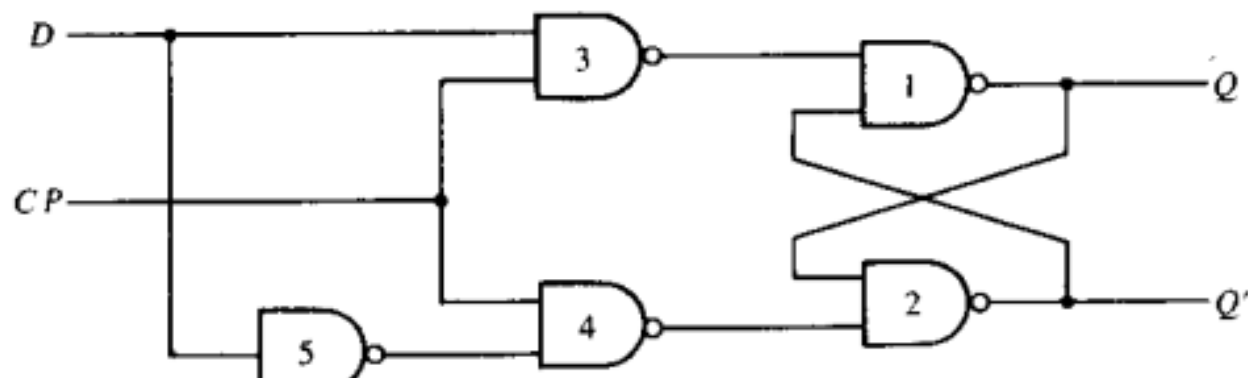
		JK		J	
		00	01	11	10
Q	0			1	1
Q	1	1			1
		K			

$Q(t+1) = JQ' + K'Q$

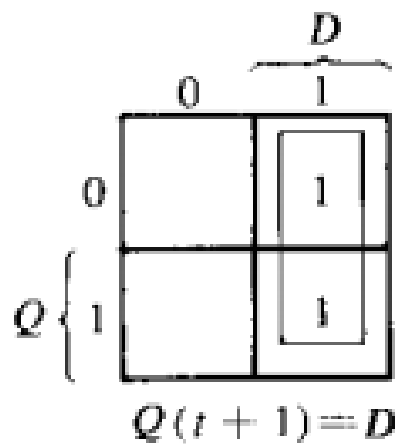
D Flip Flop

- **Modification** of the clocked RS flip-flop
- D goes **directly to S** and as **complement (inverter) to R**, reducing the number of inputs
- *If $CP = 0$* , it will be independent of S and R inputs and both will be 1 for main circuit resulting in memory state.
- If **D=1**, switch the flip-flop to the **set state**
- If **D=0**, switch the flip-flop to the **reset/clear state**
- D flip-flop receives the this designation from its ability to transfer “**data**” into flip-flop

D Flip Flop...

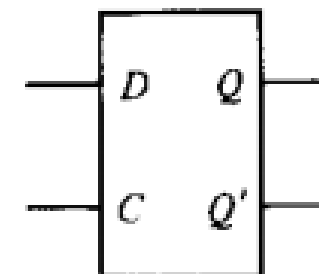


Q	D	$Q(t + 1)$
0	0	0
0	1	1
1	0	0
1	1	1



D Flip-Flop

D	$Q(t + 1)$	
0	0	Reset
1	1	Set

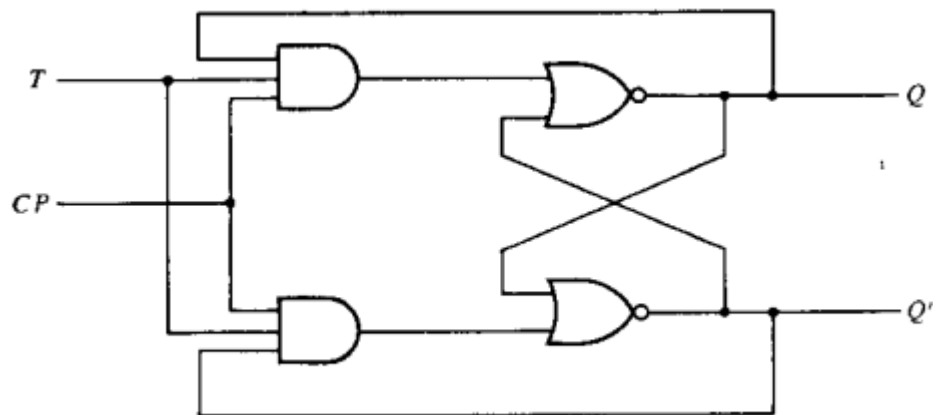




T flip flop

- Single input version of JK flip-flop
- T flip-flop is obtained from a JK flip-flop if both inputs are tied together
- The designation of T comes from the ability of the flip-flop to “toggle” or change the state
- When $T=1$, it will complement the present state ($CP=1$).

T flip flop...

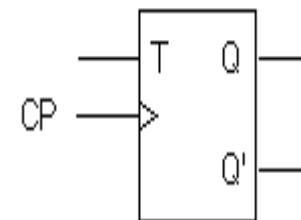


Q	T	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

	0	$\overbrace{1}^T$
0		1
$\underbrace{1}_Q$	1	

$$Q(t+1) = TQ' + T'Q$$

0	$Q(t)$	No change
1	$Q'(t)$	Complement





Administration



Administration



Administration



Administration